AP PRECALCULUS

Unit 1 • Lesson 1.1 practice • Answer key

Lesson 1.1 • Change in Tandem

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

The table gives values of f. On which interval does the table show f decreasing?

x	-2	-1	0	1	2
$f(x)$	7	2	-1	-2	-1

(A) $[-2, 1]$
(B) $[-1, 2]$
(C) $[0, 2]$
(D) $[-2, 2]$

Solution

Answer: (A). The outputs decrease from 7 to -2 as x goes from -2 to 1, then increase.

Question 2

What are the zeros of $g(x) = (x - 4)(x + 1)$?

(A) -4 and 1
(B) 4 only
(C) -1 and 4
(D) 0 and 4

Solution

Answer: (C). $g(x) = 0$ when $x - 4 = 0$ or $x + 1 = 0$.

Question 3

A function is increasing and its rate of change is decreasing. Which describes its graph?

(A) Increasing and concave up
(B) Increasing and concave down
(C) Decreasing and concave up
(D) Decreasing and concave down

Solution

Answer: (B). A decreasing rate of change means the graph is concave down.

Question 4

A car's distance d (meters) from a stop sign is recorded: $d(0) = 0$, $d(1) = 10$, $d(2) = 40$, $d(3) = 90$. Which statement fits the data?

(A) d is decreasing and concave up.
(B) d is increasing and concave down.
(C) d is linear.
(D) d is increasing and concave up.

Solution

Answer: (D). The changes $10, 30, 50$ are positive and increasing: increasing and concave up.

Question 5 (Free response)

The height h, in meters, of a ball is recorded every half second.

t (s)	0	0.5	1	1.5	2
$h(t)$ (m)	2	9	12	11	6

(a) Find the change in h over each half-second interval.
(b) Based on these changes, is the graph of h concave up or concave down on $[0, 2]$? Give a reason.
(c) Between which two recorded times does h change from increasing to decreasing, according to the table?

Solution
(a) $9 - 2 = 7$, $12 - 9 = 3$, $11 - 12 = -1$, $6 - 11 = -5$.
(b) The changes over equal intervals are decreasing ($7 > 3 > -1 > -5$), so the rate of change is decreasing: concave down.
(c) The changes go from positive ($+3$ on $[0.5, 1]$) to negative (-1 on $[1, 1.5]$), so the ball reaches its maximum height between $t = 0.5$ and $t = 1.5$; the table shows h increasing up to $t = 1$ and decreasing after it.